MATH 215 - Problem Set 05
Special Discrete Random Variables

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1. Show that:
$$E\left[X^2\right] = n^2p^2 - np^2 + np$$
for a binomial random variable with size parameter n and success probability p.

2. Show that:
$$E\left[X^2\right] = \frac{2-p}{p^2}$$
for a geometric random variable with success probability p.

3. An urn contains N white and M black balls. Balls are randomly selected, one at a time, until a black one is obtained. If we assume that each selected ball is replaced before the next one is drawn, what is the probability that:
   (a) exactly k draws are needed?
   (b) at least k draws are need?
   (c) at most k draws are needed?

4. On a given day, maintenance crews arrive at a tool crib requesting a particular spare part according to a Poisson distribution with parameter $\lambda = 2\frac{\text{arrivals}}{\text{day}}$. Three of these spare parts are normally kept on hand. If more than three orders occur, the crews must journey a considerable distance to central stores.
   (a) On a given day, what is the probability that such a journey must be made?
   (b) What is the expected demand per day for spare parts?
   (c) At least, how many spare parts must be carried if the tool crib wants to service all incoming crews at least 95% of the time?

5. Suppose that airplane engines operate independently in flight and fail with probability $q = 0.2$. Assuming that a plane makes a safe flight if at least half of its engines run, determine whether four-engine plane or a two-engine plane has the highest probability for a successful flight. Repeat the exercise for $q = \frac{1}{3}$ and $q = 0.5$.

6. In an inventory study it was determined that on the average demand for a particular item at a warehouse were made five times per day according to the Poisson distribution. What is the probability that on a given day this item is requested:
   (a) More than five times?
   (b) Not at all?

7. Suppose that on the average 1 person in 1000 make a numerical error in preparing their income tax returns. If 10000 forms are selected at random and examined, find the probability that 6, 7, or 8 of the forms will be in error. (Hint: Apply Poisson approximation to binomial random variable.)

8. Let X be a Bernoulli random variable with $E[X] = a$ and $Var[X] = b$.
   (a) Determine b in terms of a.
   (b) What is $E[Var[X]]$?
   (c) What is $Var[E[X]]$?

9. A ball is drawn from an urn containing 3 white and 3 black balls. After the ball is drawn, it is replaced and another ball is drawn. This process goes on indefinitely. What is the probability that, of the first 4 balls drawn, exactly 2 are white?

10. On a multiple-choice exam with 3 possible answers for each of the 5 questions, what is the probability that a student will get 4 or more correct answers just by guessing?

11. Pam is getting ready to take an important oral examination and is concerned about the possibility of having an "on" day or an "off" day. She figures that if she has an on day, then each of her examiners will pass her, independently of each other, with probability 0.8, whereas if she has an off day, this probability will be reduced to 0.4. Suppose that Pam will pass the examination if a majority of the examiners pass her. If Pam feels that she is twice as likely to have an off day as she is to have an on day, should she request an examination with 3 examiners or with 5 examiners?

12. When coin 1 is flipped, it lands on heads with probability 0.4; when coin 2 is flipped, it lands on heads with probability 0.7. One of these coins is randomly chosen and flipped 10 times. What is the probability that the coin lands on heads on exactly 7 of the 10 flips?

13. The expected number of typographical errors on a page of a certain magazine is 0.2. Assuming that the number of typographical errors on any page is a Poisson random variable, what is the probability that the next page you read contains:

    (a) no errors?

    (b) 2 or more typographical errors?

14. The worldwide number of airplane crashes of commercial airlines follows a Poisson distribution with an average of 0.075 crashes per month. What is the probability that there will be

    (a) at least 2 such accidents in the next month?

    (b) at most 1 accident in the next month?

15. Suppose that the average number of cars abandoned weekly on a certain highway is 2.2. Assuming a Poisson distribution for the randomness, find the probability that there will be

    (a) no abandoned cars in the next week.

    (b) more than 2 abandoned cars in the next week.

16. A certain typing agency employs 2 typists. The average number of errors per article is 3 when typed by the first typist and 4.2 when typed by the second. We assume that the number of errors per article follows independent Poisson distributions for each typist. Suppose that your article is equally likely to be typed by either typist.

    (a) Find the probability that it will have no errors.

    (b) Given that your article has no errors, what is the probability that it has been typed by the first typist?

17. Suppose that the number of accidents occurring on a highway each day is a Poisson random variable with parameter $\lambda = 3$.

    (a) Find the probability that 3 or more accidents occur today.

    (b) Repeat part (a) under the assumption that at least 1 accident occurs today.

18. If you buy a lottery ticket in 50 lotteries, in each of which your chance of winning a prize is $\frac{1}{100}$, what is the (approximate) probability that you will win a prize:

    (a) at least once?
    (b) exactly once?
    (c) at least twice?

19. The probability of being dealt a full house in a hand of poker is approximately 0.00144. Find an approximation for the probability that, in 1000 hands of poker, you will be dealt at least 2 full houses.

20. The suicide rate in a certain state is 1 suicide per 100000 inhabitants per month.

    (a) Find the approximate probability that, in a city of 400000 inhabitants within this state, there will be 8 or more suicides in a given month.
    (b) What is the probability that there will be at least 2 months during the year that will have 8 or more suicides?
    (c) Counting the present month as month number 1, what is the probability that the first month to have 8 or more suicides will be month number 5?

21. Consider a roulette wheel consisting of 38 numbers 1 through 36, 0, and double 0. If Smith always bets that the outcome will be one of the numbers 1 through 12, what is the probability that

    (a) Smith will lose his first 5 bets?
    (b) his first win will occur on his fourth bet?

22. In a call center, there are two operators at work. The first operator is more successful than the second operator and solves the problem of customers on the phone with probability 0.05 at each arriving call. This probability is equal to 0.02 for the second operator. Suppose the manager randomly selects one of these operators without knowing which one is more successful, what is the probability that:

    (a) exactly one of the next five customers' problems will be solved on the phone?
    (b) the first customer problem that is solved on the phone occurs in the fourth telephone call?
    (c) exactly five of the next 200 customers' problems will be solved on the phone? (Hint: Use a Poisson approximation.)
    (d) If the first customer problem that is solved on the phone occurs in the fourth telephone call, what is the probability that the second operator is selected by the manager?

23. From a lot of 10 missiles, 4 are selected at random and fired. If the whole lot contains 3 defective missiles that will not fire, what is the probability that

    (a) All 4 will fire?
    (b) At most 2 will not fire?

24. An annexation suit is being considered against a county subdivision of 1200 residences by a neighboring city. If the occupants of half the residences object to being annexed, what is the probability that in a random sample of 10 at least 3 favor the annexation suit? (Hint: Use binomial approximation on the hypergeometric RV.)

25. To avoid detection at customs, a traveler has placed six narcotic tablets in a bottle containing nine vitamin pills that are similar in appearance. If the customs official selects three of the tablets at random for analysis, what is the probability that the traveler will be arrested for illegal possession of narcotics?

26. It is estimated that 4000 of the 10000 voting residents of a town are against a new sales tax. If 15 eligible voters are selected at random and asked their opinion, what is the probability that at most 7 favor the new tax?

27. Suppose that each birth that a mother bunny gives is a female with probability 1/3 independent from the number of births that she can give.

    (a) What is the probability that the 3rd female is born on the 8th birth?

    (b) If X denotes the number of births that a mother bunny has to give in order to have a 3rd female, what is $E[X]$?

    (c) If a mother bunny has 3 female and 6 male offsprings and we choose 4 of them randomly, what is the probability that we will have at least two but at most three males?

28. The paper company, where James and Dwight are employed as salesperson, decides to start a promotional campaign for its clients. Suppose the number of clients is sufficiently large. Their decision is to give a type I promotion to a randomly chosen five of their clients and type II promotion to a randomly chosen five of their remaining clients. 60 percent of the company clients are owned by Dwight while the 40 percent of the clients are owned by James. In order to distinguish the clients who win type I and type II promotions, the company first draws 10 from their clients randomly. Then from this group of 10, they choose five clients to award with type I promotion, so the remaining five can get the type II promotion.

    (a) In the chosen 10, what is the probability that Dwight has 6 and James has 4 clients?

    (b) Given that there are 4 clients of James and 6 clients of Dwight are present in the chosen 10, what is the probability that exactly one of James's clients and four of Dwight's clients are awarded with type II promotion?

    (c) Assume we restart the drawing procedure. Let the random variable X denote the number of James's clients who wins a type I promotion.

        i. What kind of random variable is X? What is/are its parameter/parameters?

        ii. What is $E[X]$?

        iii. What is $Var[X]$?

29. The number of misprints in a randomly selected page of a certain book follows the Poisson distribution with an average of 0.25 misprints per page. If we assume independence between the pages:

    (a) what is the probability of having no misprints in exactly three of the next five pages?

    (b) starting from page 1, what is the probability that the first page that has at least one misprint is the 6th page?

30. In a national park, there are 813 elks, 475 of which being female. We call the elks, one by one and randomly, for health inspection. Inspected elks are moved to a barn afterwards, independent of the gender. We continue this process until four females are collected in the barn. What is the probability that there are 8 elks in the barn? (Hint: You may apply an approximation based on the theory.)

31. Past statistics show that, on the average 1 out of 500 people can answer a certain conundrum correctly. We ask this conundrum to 1500 people, independently from each other. It is known that there has been at least two people to give a correct answer. Conditioned on that fact, what is the probability that we have got at least four correct answers? (Hint: You may apply an approximation based on the theory.)

32. A certain town has 37950 residents, 12250 of which are children below age 18. Among these residents we randomly select a group of 12 people. What is the probability that exactly half of the group to consist of children?

33. Pam is a professional basketball player. She is likely to score a 3-pts shot, independently at each trial, with 0.6 probability if she is on an "on" day. However, this score probability reduces to 0.35 on an "off"day. Luckily, she is on an "on" day in 80% of the games she plays.

    (a) In a random game on a random day, what is the probability that her third 3-pts score in the game will be in her sixth 3-pts shot?

    (b) If Pam scores her scores her third 3-pts score in her sixth 3-pts shot, what is the probability that she is on an "off" day?

34. We are shooting arrows at a target and we know that the bull's eye probability is 0.5 at each trial, independently. In a single round, we fire a total of 10 shots and, according to the results, we will be assigned with the following points:

    - 0 points, if we have no bullseye hit,
    - 1 point, if we have at least one but less than three bullseye hits,
    - 3 points, if we have at least three but less eight bullseye hits,
    - 5 points, if we have 8 or 9 bullseye hits,
    - 10 points, if we hit the bullseye in all shots.

    Let X denote the points that we gain in a single round. What is $P(1 < X \leq 8)$?

35. Consider the experiment in which we flip an unfair coin, with $\frac{1}{3}$ tail probability, until we observe a tail. 8 students, each owning such a coin, are independently performing this experiment. What is the probability that exactly half of the students to finish the experiment exactly in three flips?

36. Past statistics show that, on the average 2 out of 1000 people can answer a certain conundrum correctly. We ask this conundrum to 1200 people, independently from each other. What is the probability to get at least three correct answers? Choose the closest value.

37. The monthly number of fire incidents observed in Cekmekoy is a Poisson random variable with an average of 2.4 fires per month. Suppose all months consist of 30 days, the incidents occur independently in all months, and we are in the beginning of a year. What is the probability that the first month with no fire incident will be April?

38. The number of times that a person experiences flu infection in a year is a Poisson random variable with an average of 1.5 infections per year. A certain vaccine reduces this infection rate to 0.5 infections per year, without changing the probability distribution. We know also 25% of the population is shot with that vaccine. If a randomly selected person experiences flu exactly ones in the next year, what is the probability that this person was vaccinated.

39. James is a big fan of "Seinfeld" comedy series and on any day, the number of "Seinfeld" episodes he watches follows a Poisson distribution with an average of 3 episodes per day. When we ask James if he has watched any episodes of Seinfeld yesterday, he answers positively. Then, what is the probability that he has watched more than one episode yesterday?

40. A deck of cards contains 20 cards: 6 red cards and 14 blue cards. 5 cards are drawn randomly without replacement.

    (a) What is the probability that exactly 3 red cards are drawn?

    (b) Let X denote the number of red cards drawn. Calculate the probability mass function and the expected value of X.

Solutions

1. We can express x^2 as $x(x-1)+x$ and calculate $E\left[X^2\right]$ as follows:

$$\begin{aligned}
E\left[X^2\right] &= \sum_{x=0}^{n} x^2 \begin{pmatrix} n \\ x \end{pmatrix} p^x (1-p)^{n-x} \\
&= \sum_{x=0}^{n} x(x-1) \tfrac{n!}{x!(n-x)!} p^x (1-p)^{n-x} + \underbrace{\sum_{x=0}^{n} x \frac{n!}{x!\,(n-x)!} p^x (1-p)^{n-x}}_{E[X]=np} \\
&= n(n-1)p^2 \left(\sum_{x=2}^{n} \tfrac{(n-2)!}{(x-2)!(n-x)!} p^{x-2} (1-p)^{n-x} \right) + np \\
&= n(n-1)p^2 \underbrace{\left(\sum_{i=0}^{n-2} \frac{(n-2)!}{i!\,(n-i-2)!} p^i (1-p)^{n-i-2} \right)}_{1} + np \\
&= n^2p^2 - np^2 + np
\end{aligned}$$

2. We can express x^2 as $x(x+1)-x$ and calculate $E\left[X^2\right]$ as follows:

$$\begin{aligned}
E\left[X^2\right] &= \sum_{x=1}^{\infty} x^2 pq^{x-1} = \sum_{x=1}^{\infty} (x+1)\, xpq^{x-1} - \underbrace{\sum_{x=1}^{\infty} xpq^{x-1}}_{E[X]=\frac{1}{p}} \\
&= -\tfrac{1}{p} + \sum_{x=1}^{\infty} p \frac{\partial^2 q^{x+1}}{\partial q^2} = -\tfrac{1}{p} + p \frac{\partial^2 \sum_{x=1}^{\infty} q^{x+1}}{\partial q^2} \\
&= -\tfrac{1}{p} + p \frac{\partial^2 \left(q^2+q^3+q^4+q^5+\ldots\right)}{\partial q^2} \\
&= -\tfrac{1}{p} + p \frac{\partial^2 \left(\frac{q^2+q-1}{1-q}\right)}{\partial q^2} \\
&= -\tfrac{1}{p} + p \frac{\partial \left(\frac{2q-q^2}{(1-q)^2}\right)}{\partial q} \\
&= -\tfrac{1}{p} + \frac{2p^4+2p^2\left(1-p^2\right)}{p^4} = \tfrac{2-p}{p^2}
\end{aligned}$$

3. This is a sequence of Bernoulli trials for which success is defined as drawing a black ball. The success probability is $p = \frac{M}{M+N}$ and the failure probability is $q = 1-p = \frac{N}{M+N}$. Say, X is the trial where we observe the first success, i.e., a geometric random variable.

$$X \sim Geom\left(p = \frac{M}{M+N}\right)$$

   (a) We know that the first success occurs at the kth draw with probability:

$$P\{X=k\} = pq^{k-1} = \tfrac{MN^{k-1}}{(M+N)^k}$$

   (b)

$$\begin{aligned}
P\{X \geq k\} &= \sum_{x=k}^{\infty} P\{X=x\} = \sum_{x=k}^{\infty} pq^{x-1} = p\left(q^{k-1}+q^k+q^{k+1}+\ldots\right) \\
&= pq^{k-1} \underbrace{\left(1+q+q^2+\ldots\right)}_{\frac{1}{1-q}} = \tfrac{pq^{k-1}}{1-q} = q^{k-1}
\end{aligned}$$

$$P\{X \geq k\} = q^{k-1} = \left(\frac{N}{M+N}\right)^{k-1}$$

(c)

$$P\{X \leq k\} = 1 - P\{X \geq k+1\} = 1 - q^k = 1 - \left(\frac{N}{M+N}\right)^k$$

4. Let X denote the number of crew arrivals on any day. The question tells us that:

$$X \sim Pois\,(\lambda = 2)$$

(a)

$$\begin{aligned} P(X > 3) &= 1 - P(X \leq 3) \\ &= 1 - \sum_{x=0}^{3} P(X = x) \\ &= 1 - \sum_{x=0}^{3} e^{-2}\tfrac{2^x}{x!} \\ &= 1 - e^{-2}\left(\tfrac{2^0}{0!} + \tfrac{2^1}{1!} + \tfrac{2^2}{2!} + \tfrac{2^3}{3!}\right) \\ &= 1 - \tfrac{19}{3}e^{-2} \approx 0.1429 \end{aligned}$$

(b) For a Poisson random variable, the expected value is equal to its rate parameter. Therefore $E[X] = \lambda = 2$ per day.

(c) We are interested in the minimal x value for which $F(x) = P(X \leq x) \geq 0.95$. We can find this x with trial and error.

$$P(X \leq 3) = \sum_{x=0}^{3} e^{-2}\frac{2^x}{x!} = e^{-2}\left(\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!}\right) \approx 0.8571$$

$$P(X \leq 4) = \sum_{x=0}^{4} e^{-2}\frac{2^x}{x!} = e^{-2}\left(\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!}\right) \approx 0.9473$$

$$P(X \leq 5) = \sum_{x=0}^{5} e^{-2}\frac{2^x}{x!} = e^{-2}\left(\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!} + \frac{2^5}{5!}\right) \approx 0.9834$$

So we need to keep at least $x = 5$ parts, so that the tool crib serves all incoming crews at least 95% of the time. Please note that, since X is a discrete random variable $P(X \leq x) = 0.95$ may not have a solution. This is why we were looking for an x that satisfies $P(X \leq x) \geq 0.95$.

5. The probability for any engine to complete a flight without any failure is:

$$p = 1 - q = 1 - 0.2 = 0.8$$

Let X be the number of engines that will complete a flight without a failure in a four-engine plane:

$$X \sim Binom\,(n = 4, p = 0.8)$$

For a four-engine plane to complete a flight, the number of engines that will run without a failure should be at least 2. Hence, the successful flight probability for a four-engine plane is:

$$P(X \geq 2) = \sum_{x=2}^{4} P(X = x) = \sum_{x=2}^{4} \binom{4}{x} p^x (1-p)^{4-x}$$

Let Y be the number of engines that will complete a flight without a failure in a two-engine plane:

$$Y \sim Binom\,(n = 2, p = 0.8)$$

For a two-engine plane to complete a flight, the number of engines that will run without a failure should be at least 1. Hence, the successful flight probability for a two-engine plane is:

$$P(Y \geq 1) = \sum_{y=1}^{2} P(Y = y) = \sum_{y=1}^{2} \binom{2}{y} p^y (1-p)^{2-y}$$

We can use these formulas to fill the following table, each row of which compares the successful flight probability of a four-engine plane to a two-engine plane. The last column provides the results.

		Successful Flight Probability		
Eng. Fail Prob. (q)	Eng. No Fail Prob. (p)	4-Eng.	2-Eng.	Choice
$q = 0.2$	$p = 0.8$	0.9728	0.96	4-Engines
$q = \frac{1}{3}$	$p = \frac{2}{3}$	$\frac{8}{9}$	$\frac{8}{9}$	Both
$q = 0.5$	$p = 0.5$	0.6875	0.75	2-Engines

6. Let X denote the daily demand to the item, which is Poisson distributed with rate parameter $\lambda = 5$.

$$X \sim Pois\left(\lambda = 5\frac{\text{units}}{\text{day}}\right)$$

(a) The question ask $P(X > 5)$.

$$\begin{aligned} P(X > 5) &= 1 - P(X \leq 5) \\ &= 1 - \sum_{x=0}^{5} P(X = x) \\ &= 1 - \sum_{x=0}^{5} e^{-5}\frac{5^x}{x!} \\ &= 1 - e^{-5}\left(\frac{5^0}{0!} + \frac{5^1}{1!} + \frac{5^2}{2!} + \frac{5^3}{3!} + \frac{5^4}{4!} + \frac{5^5}{5!}\right) \\ &= 1 - e^{-5}\left(1 + 5 + \frac{25}{2} + \frac{125}{6} + \frac{625}{24} + \frac{625}{24}\right) \\ &= 1 - \frac{1097}{12}e^{-5} \approx 0.3840 \end{aligned}$$

(b) The question ask $P(X = 0) = e^{-5}\frac{5^0}{0!} = e^{-5} \approx 0.0067$.

7. The probability of a person to make a numerical error in preparing their income tax returns is $p = \frac{1}{1000} = 0.001$ Assuming independence between the tax forms, out of 10000 forms, the number of forms that will be in error is a binomial random variable:

$$X \sim Binom(n = 10000, p = 0.001)$$

The question asks the probabilities $P(X = x)$ for $x = 6, 7, 8$. Though a computer software can successfully calculate these probabilities, manual methods can yield numerical errors. We know that Poisson distribution is a limiting case of binomial distribution when the size parameter n is large and success probability is quite small. Hence, we can make the following approximation:

$$X \approx X' \sim Pois(\lambda = np = 10000 \times 0.001 = 10)$$

The probabilities can be approximately through Poisson probability mass function:

$$P(X = 6) \approx P(X' = 6) = e^{-10}\frac{10^6}{6!} \approx 0.0631$$

$$P(X = 7) \approx P(X' = 7) = e^{-10}\frac{10^7}{7!} \approx 0.0901$$

$$P(X = 8) \approx P(X' = 8) = e^{-10}\frac{10^8}{8!} \approx 0.1126$$

which are quite close to exact probabilities calculated with binomial probability mass function.

8. Let $X \sim Bern(p)$. The question tells us that:

$$E[X] = p = a, \qquad Var[X] = p(1-p) = b$$

.

(a) Then $b = a(1-a)$.

(b) $Var[X] = b$ is a constant value. The expected value of a constant (non-random) value is equal to itself. Thus:

$$E[Var[X]] = E[b] = b$$

(c) $E[X] = a$ is a constant value. The variance of a constant (non-random) value is zero. Thus:

$$Var[E[X]] = Var[a] = 0$$

9. When the drawn balls are replaced before the next draw, the draws will become independent trials. At each time, the probability of selecting a white ball is $p = \frac{3}{6} = 0.5$. The question examines the first 4 trials. Let X denote the number of white balls selected in the first four trials:

$$X \sim Binom(n = 4, p = 0.5)$$

and the probability of interest is:

$$P(X = 2) = \binom{4}{2}(0.5)^2(1-0.5)^{4-2} = 0.375$$

10. At each guess, student is likely to choose the correct answer with $p = \frac{1}{3}$ probability. Assuming independence between guesses, the number of correct answers the student will have out of five guesses is binomially distributed.

$$X \sim Binom\left(n = 5, p = \frac{1}{3}\right)$$

The question asks the following probability:

$$\begin{aligned} P(X \geq 4) &= P(X = 4) + P(X = 5) \\ &= \binom{5}{4}\left(\tfrac{1}{3}\right)^4\left(\tfrac{2}{3}\right)^1 + \binom{5}{5}\left(\tfrac{1}{3}\right)^5\left(\tfrac{2}{3}\right)^0 \\ &= \tfrac{10}{243} + \tfrac{1}{243} \\ &= \tfrac{11}{243} \end{aligned}$$

11. This question requires a knowledge on conditional probability as well as a knowledge on the binomial distribution. Let E denote the event that Pam is on an "on" day. The question provides that:

$$P(E) = \tfrac{1}{3}, \quad P(E^c) = \tfrac{2}{3}$$

The question also provides that, the number of examiners that will pass Pam is a binomial random variable with a conditional success probability on the fact that she is on an "on" or "off" day.

$$X|E \sim Binom(n, p = 0.8), \quad X|E^c \sim Binom(n, p = 0.4)$$

Pam will pass the exam if $X > \frac{n}{2}$. Thus we need to maximize $P\{X > \frac{n}{2}\}$.

If $n = 3$:

$$\begin{aligned} P\{X > 1.5\} &= P(E)\,P\{X > 1.5|E\} + P(E^c)\,P\{X > 1.5|E^c\} \\ &= \tfrac{1}{3}\sum_{x=2}^{3}\binom{3}{x}(0.8)^x(0.2)^{3-x} + \tfrac{2}{3}\sum_{x=2}^{3}\binom{3}{x}(0.4)^x(0.6)^{3-x} \approx 0.5333 \end{aligned}$$

If $n = 5$:

$$\begin{aligned} P\{X > 2.5\} &= P(E)\,P\{X > 2.5|E\} + P(E^c)\,P\{X > 2.5|E^c\} \\ &= \tfrac{1}{3}\sum_{x=3}^{5}\binom{5}{x}(0.8)^x(0.2)^{5-x} + \tfrac{2}{3}\sum_{x=3}^{5}\binom{5}{x}(0.4)^x(0.6)^{5-x} \approx 0.5257 \end{aligned}$$

So, taking the exam with 3 examiners is a better choice for Pam.

12. Let A denote the event of selecting coin 1. Obviously:

$$P(A) = P(A^c) = 0.5$$

Let X denote the number of heads observed in 10 independent coin flips. The distribution of X is not known. However, it is known conditionally on the coin that is selected.

$$X|A \sim Binom\,(n = 10, p = 0.4)\,, \quad X|A^c \sim Binom\,(n = 10, p = 0.7)$$

The question asks $P(X = 7)$, which can be calculated by conditioning on the coin type.

$$\begin{aligned} P(X = 7) &= P(A)\,P(X = 7|A) + P(A^c)\,P(X = 7|A^c) \\ &= 0.5\binom{10}{7}(0.4)^7(0.6)^3 + 0.5\binom{10}{7}(0.7)^7(0.3)^3 \\ &\approx 0.5 \times 0.0425 + 0.5 \times 0.2668 \\ &\approx 0.1546 \end{aligned}$$

13. Let X denote the number of typographical errors on the next page. We know that $E[X] = 0.2$, thus:

$$X \sim Pois\left(\lambda = 0.2\frac{\text{error}}{\text{page}}\right)$$

(a) $P(X = 0) = e^{-0.2}\frac{(0.2)^0}{0!} \approx 0.8187$

(b) $P(X \geq 2) = 1 - P(X < 2) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-0.2}\frac{(0.2)^0}{0!} - e^{-0.2}\frac{(0.2)^1}{1!} \approx 0.0175$

14. Let X denote the worldwide number of airplane crashes of commercial airlines in the next month. We know that $E[X] = 0.075$, thus:

$$X \sim Pois\left(\lambda = 0.075\frac{\text{accident}}{\text{month}}\right)$$

(a) $P(X \geq 2) = 1 - P(X < 2) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-0.075}\frac{(0.075)^0}{0!} - e^{-0.075}\frac{(0.075)^1}{1!} \approx 0.0027$

(b) $P(X \leq 1) = 1 - P(X \geq 2) \approx 1 - 0.0027 = 0.9973$

15. Let X denote the number of cars abandoned in the next week. We know that $E[X] = 2.2$, thus:

$$X \sim Pois\left(\lambda = 2.2\frac{\text{car}}{\text{week}}\right)$$

(a) $P(X=0) = e^{-2.2}\frac{(2.2)^0}{0!} \approx 0.1108$

(b) $P(X>2) = 1 - P(X=0) - P(X=1) - P(X=2) = 1 - e^{-2.2}\frac{(2.2)^0}{0!} - e^{-2.2}\frac{(2.2)^1}{1!} - e^{-2.2}\frac{(2.2)^2}{2!} \approx 0.3773$

16. Let A denote the event that the article is written by the first typist. It is stated by the question that:

$$P(A) = P(A^c) = 0.5$$

Let X denote the number of errors observed in the article (for which we do not know the typist). The distribution of X is not known. However, it is known conditionally on the typist.

$$X|A \sim Pois\left(\lambda = 3\frac{\text{error}}{\text{article}}\right), \quad X|A^c \sim Pois\left(\lambda = 4.2\frac{\text{error}}{\text{article}}\right)$$

(a) The question asks $P(X=0)$, which can be calculated by conditioning on the typist.

$$\begin{aligned} P(X=0) &= P(A)P(X=0|A) + P(A^c)P(X=0|A^c) \\ &= 0.5e^{-3}\frac{3^0}{0!} + 0.5e^{-4.2}\frac{(4.2)^0}{0!} \\ &\approx 0.0324 \end{aligned}$$

(b) This is basically a question that is solved by Bayes' formula. However, half of the information is already generated in part (a).

$$\begin{aligned} P(A|X=0) &= \frac{P(A\cap(X=0))}{P(X=0)} \\ &= \frac{P(A)P(X=0|A)}{P(A)P(X=0|A)+P(A^c)P(X=0|A^c)} \\ &= \frac{0.5e^{-3}\frac{3^0}{0!}}{0.5e^{-3}\frac{3^0}{0!}+0.5e^{-4.2}\frac{(4.2)^0}{0!}} \approx \frac{0.0249}{0.0324} \approx 0.7685 \end{aligned}$$

17. Let X denote the number of accidents that will occur today on the highway:

$$X \sim Pois\left(\lambda = 3\frac{\text{accident}}{\text{day}}\right)$$

(a) $P(X \geq 3) = 1 - \sum_{x=0}^{2} e^{-3}\frac{3^x}{x!} = 1 - e^{-3}\left(\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!}\right) = 1 - \frac{17}{2}e^{-3} \approx 1 - 0.4232 = 0.5768$

(b) The question asks a conditional probability here. It asks:

$$\begin{aligned} P\{X \geq 3|X \geq 1\} &= \frac{P\{(X\geq 3)\cap(X\geq 1)\}}{P\{X\geq 1\}} = \frac{P\{X\geq 3\}}{P\{X\geq 1\}} \\ &= \frac{1-P\{X=0\}-P\{X=1\}-P\{X=2\}}{1-P\{X=0\}} \approx \frac{0.5768}{0.9502} \approx 0.6070 \end{aligned}$$

18. Assuming independence between lottery tickets, the number of winning tickets out of 50 is a binomial random variable, which can be approximated by a Poisson random variable since the size parameter n is moderately large and the success probability p is small.

$$X \sim Binom\,(n = 50, p = 0.01) \quad \sim \quad X' \sim Pois\,(\lambda = np = 0.05)$$

It may not be a good approximation since the parameters are not too extreme. But the question is telling us to do so:

(a) $P\,(X' \geq 1) = 1 - P\,(X' = 0) = 1 - e^{-0.05}\frac{(0.05)^0}{0!} \approx 0.3935$

(b) $P\,(X' = 1) = e^{-0.05}\frac{(0.05)^1}{1!} \approx 0.3033$

(c) $P\,(X' \geq 2) = 1 - P\,(X' = 0) - P\,(X' = 1) = 1 - e^{-0.05}\frac{(0.05)^0}{0!} - e^{-0.05}\frac{(0.05)^1}{1!} \approx 0.0902$

Note that, these answers would be 0.3950, 0.3056, and 0.0894, if we have used the binomial probability mass function and a computer software, which shows that the approximations are not very good and $n = 50$ is not a very large size parameter to follow a Poisson approximation to binomial random variable.

19. A full house is a hand that contains three of a kind of card with another two of a kind in five cards. Of a 52-deck with 13 kinds, the probability of getting a full house directly from the deal is:

$$\frac{2\binom{13}{2}\binom{4}{3}\binom{4}{2}\binom{44}{0}}{\binom{52}{5}} = \frac{3744}{2598960} \approx 0.00144$$

But this value was already given by the question, so we have just re-calculated it for fun. Now, the number of full houses obtained directly from the deal out of 1000 hands is a binomial random variable (say X), which should be approximated by a Poisson random variable (say $X\prime$) as the size parameter $n = 1000$ is extremely large and the success probability $p = 0.00144$ is extremely small.

$$X \sim Binom\,(n = 1000, p = 0.00144) \quad \sim \quad X' \sim Pois\,(\lambda = np = 1.44)$$

The question asks for the approximate probability:

$$P\,(X' \geq 2) = 1 - P\,(X' = 0) - P\,(X' = 1) = 1 - e^{-1.44}\frac{(1.44)^0}{0!} - e^{-1.44}\frac{(1.44)^1}{1!} \approx 0.4219$$

The exact value of this probability can be found as 0.4220 by using the binomial probability mass function and a computer software, which shows that the approximation is very good.

20. In the state, the probability of an inhabitant to commit suicide in the following month can be assumed as $p = \frac{1}{100000} = 0.00001$. Unless there is a cult of such, we can assume independence between the inhabitants' intention for suicide.

(a) In the following month, the number of suicides in a city of 400000 inhabitants (say X) would be binomial distributed. However we can apply a Poisson approximation since the size parameter $n = 400000$ is extremely large and the individual suicide probability $p = 0.00001$ is extremely small.

$$X \sim Binom\,(n = 400000, p = 0.00001) \quad \sim \quad X' \sim Pois\,(\lambda = np = 4)$$

The question asks for $P\,(X \geq 8)$, so let us approximate it:

$$P\,(X' \geq 8) = 1 - \sum_{x=0}^{7} P\,(X' = x)1 - \sum_{x=0}^{7} e^{-4}\frac{4^x}{x!} \approx 0.0511$$

(b) In a year we have 12 months, and the probability in any month to have 8 or more suicides is calculated as 0.0511 in the previous part. We assume independence between the months. Out of 12 months, the number of months in which we will have 8 or more suicides is another binomial random variable (say Y).

$$Y \sim Binom\,(n = 12, p = 0.0511)$$

and the probability of interest is $P\,(Y \geq 2)$, which can be calculated directly.

$$\begin{aligned} P\,(Y \geq 2) &= 1 - P\,(Y = 0) - P\,(Y = 1) \\ &= 1 - \begin{pmatrix} 12 \\ 0 \end{pmatrix} (0.0511)^0\,(0.9489)^{12} - \begin{pmatrix} 12 \\ 1 \end{pmatrix} (0.0511)^1\,(0.9489)^{11} \\ &\approx 1 - 0.5329 - 0.3444 \\ &= 0.1227 \end{aligned}$$

At the point you should notice two things:

- The probability that is calculated by a probability distribution can as well be a parameter to another probability distribution as in this example. That is a very ordinary thing to observe in real life.
- The probability 0.0511 found in part (a) was approximated to four decimal digits, which caused a loss of information when we were calculating the result in part (b). Making too much approximation by rounding decimal digits at each step will cause trouble in the end. If we have used a computer software (which does not round the results), the answer to part (b) would be 0.1229. This is not too far from the solution we have found, since we have stayed conservative and rounded the probabilities to 4 decimal digits at all times.

(c) In any month to have 8 or more suicides is calculated as 0.0511 in part (a). We assume independence between the months and the first month to observe such an event would be a geometric random variable (say T):

$$T \sim Geom\,(p = 0.0511)$$

and the question asks $P\,(T = 5)$.

$$P\,(T = 5) = (0.0511)\,(1 - 0.0511)^{5-1} \approx 0.0414$$

21. Assuming that the roulette wheel is fair, in each bet, independently, Smith will win with $p = \frac{12}{38} = \frac{6}{19}$ and lose with $q = 1 - p = \frac{13}{19}$ probability.

(a) Since wins are independent, losses are independent as well. The direct answe to here is simply $\left(\frac{13}{19}\right)^5 \approx 0.1500$. However, we can also answer this indirectly.

Let X denote the number of games that Smith needs to play in order to get his first win. X is a geometric random variable with success probability $p = \frac{6}{19}$. The event that he loses his first 5 bets is equivalent to the event that he needs to play at least 6 games to get his first win. Hence:

$$\begin{aligned} P\,(X \geq 6) &= 1 - P\,(X < 6) \\ &= 1 - P\,(X = 1) - P\,(X = 2) - P\,(X = 3) - P\,(X = 4) - P\,(X = 5) \\ &= 1 - \tfrac{6}{19} - \left(\tfrac{6}{19}\right)\left(\tfrac{13}{19}\right) - \left(\tfrac{6}{19}\right)\left(\tfrac{13}{19}\right)^2 - \left(\tfrac{6}{19}\right)\left(\tfrac{13}{19}\right)^3 - \left(\tfrac{6}{19}\right)\left(\tfrac{13}{19}\right)^4 \\ &\approx 0.1500 \end{aligned}$$

(b) $P\,(X = 4) = \left(\frac{6}{19}\right)\left(\frac{13}{19}\right)^3 = 0.1012$

22. Let A denote the event that the successful operator is selected by the manager. It is stated by the question that:

$$P(A) = P(A^c) = 0.5$$

(a) Among the next 5 customers, let X denote the number of customers whose problem is solved by the operator. The distribution of X is not known. However, it is known conditionally on the selected operator.

$$X|A \sim Binom(n = 5, p = 0.05), \quad X|A^c \sim Binom(n = 5, p = 0.02)$$

The question asks $P(X = 1)$, which can be calculated by conditioning on the selected operator.

$$\begin{aligned} P(X = 1) &= P(A)P(X = 1|A) + P(A^c)P(X = 1|A^c) \\ &= 0.5\begin{pmatrix} 5 \\ 1 \end{pmatrix}(0.05)^1(0.95)^4 + 0.5\begin{pmatrix} 5 \\ 1 \end{pmatrix}(0.02)^1(0.98)^4 \\ &\approx 0.1479 \end{aligned}$$

(b) Let Y denote the number of customers until we get a problem solved. The distribution of Y is not known. However, it is known conditionally on the selected operator.

$$Y|A \sim Geom(p = 0.05), \quad Y|A^c \sim Geom(p = 0.02)$$

The question asks $P(Y = 4)$, which can be calculated by conditioning on the selected operator.

$$\begin{aligned} P(Y = 4) &= P(A)P(Y = 4|A) + P(A^c)P(Y = 4|A^c) \\ &= 0.5(0.05)(0.95)^3 + 0.5(0.02)(0.98)^3 \\ &\approx 0.0308 \end{aligned}$$

(c) Just like in part (a), among the next 200 customers, let T denote the number of customers whose problem is solved by the operator. The distribution of T is not known. However, it is known conditionally on the selected operator.

$$T|A \sim Binom(n = 200, p = 0.05), \quad T|A^c \sim Binom(n = 200, p = 0.02)$$

We assume that, under both conditions (A and A_c, for both of the operators), the size parameters are large and success probabilities small enough to apply a Poisson approximation (though, these approximations will not be good enough because 0.05 and 0.02 are not extremely but moderately small).

$$T'|A \sim Pois(\lambda = np = 10), \quad T'|A^c \sim Pois(\lambda = np = 4)$$

The question asks $P(T = 5)$, which can be approximated by $P(T' = 1)$ and can be calculated by conditioning on the selected operator.

$$\begin{aligned} P(T' = 5) &= P(A)P(T' = 5|A) + P(A^c)P(T' = 5|A^c) \\ &= 0.5e^{-10}\tfrac{10^5}{5!} + 0.5e^{-4}\tfrac{4^5}{5!} \\ &\approx 0.0971 \end{aligned}$$

(d) This is basically a question that is solved by Bayes' formula. However, half of the information is already generated in part (b).

$$\begin{aligned} P(A^c|Y = 4) &= \tfrac{P(A^c \cap (Y=4))}{P(Y=4)} \\ &= \tfrac{P(A^c)P(Y=4|A^c)}{P(A)P(Y=4|A)+P(A^c)P(Y=4|A^c)} \\ &= \tfrac{0.5(0.02)(0.98)^3}{0.5(0.05)(0.95)^3+0.5(0.02)(0.98)^3} \approx \tfrac{0.0094}{0.0308} \approx 0.3052 \end{aligned}$$

23. The number of missiles working in the selected sample (say X) is a hypergeometric random variable.

$$X \sim HypGeom\left(N = 10, K = 7, n = 4\right)$$

(a) $P\left(X = 4\right) = \dfrac{\binom{7}{4}\binom{3}{0}}{\binom{10}{4}} = \frac{35}{210} = \frac{1}{6}$

(b) The event that at most two will not fire is equivalent to the event that at least two will fire.

$$\begin{aligned} P\left(X \geq 1\right) &= P\left(X = 2\right) + P\left(X = 3\right) + P\left(X = 4\right) \\ &= \frac{\binom{7}{2}\binom{3}{2}}{\binom{10}{4}} + \frac{\binom{7}{3}\binom{3}{1}}{\binom{10}{4}} + \frac{\binom{7}{4}\binom{3}{0}}{\binom{10}{4}} \\ &= \tfrac{63}{210} + \tfrac{105}{210} + \tfrac{35}{210} = \tfrac{203}{210} = \tfrac{29}{30} \end{aligned}$$

24. Out of 1200 residences, 600 of them object to being annexed while 600 of them favor the annexation suit. The number of residences that favor the annexation in the random sample of 10 residences is a hypergeometric random variable.

$$X \sim HypGeom\left(N = 1200, K = 600, n = 10\right)$$

and the question asks $P\left(X \geq 3\right)$:

$$\begin{aligned} P\left(X \geq 3\right) &= 1 - P\left(X = 0\right) - P\left(X = 1\right) - P\left(X = 2\right) \\ &= 1 - \frac{\binom{600}{0}\binom{600}{10}}{\binom{1200}{10}} - \frac{\binom{600}{1}\binom{600}{9}}{\binom{1200}{10}} - \frac{\binom{600}{2}\binom{600}{8}}{\binom{1200}{10}} \\ &\approx 0.9460 \end{aligned}$$

However, this value is calculated by the help of a computer software due to very large values observed in the numerators and denominators. Manually, it is a good idea to use binomial approximation to get this answer (and also it is told so by the question). We know that the sample is selected without replacement, therefore there is a dependence between each selected residence. However, this dependence is not too strong since N and K are very large compared to n. The binomial approximation to hypergeometric random variable is applied as follows:

$$X \sim HypGeom\left(N = 1200, K = 600, n = 10\right) \quad \approx \quad X' \sim Binom\left(n = 10, p = \frac{K}{N} = \frac{600}{1200} = 0.5\right)$$

Thus, we calculate the same probability with the binomial approximation.

$$\begin{aligned} P\left(X' \geq 3\right) &= 1 - P\left(X' = 0\right) - P\left(X' = 1\right) - P\left(X' = 2\right) \\ &= 1 - \binom{10}{0}\left(\tfrac{1}{2}\right)^0\left(\tfrac{1}{2}\right)^{10} - \binom{10}{1}\left(\tfrac{1}{2}\right)^1\left(\tfrac{1}{2}\right)^9 - \binom{10}{2}\left(\tfrac{1}{2}\right)^2\left(\tfrac{1}{2}\right)^8 \\ &\approx 1 - \tfrac{1}{1024} - \tfrac{10}{1024} - \tfrac{45}{1024} = \tfrac{968}{1024} \approx 0.9453 \end{aligned}$$

The error size of this approximation can be criticized, yet, without a strong tool for computation, that is the only solution we will be able to get.

25. Let X be the number of narcotic tablets that the customs official will find in the selected 3 tablets. X is a hypergeometric random variable:

$$X \sim HypGeom\left(N = 15, K = 6, n = 3\right)$$

The event of an arrest is equivalent to finding at least one narcotic tablet in the selected three:

$$P\left(X \geq 1\right) = 1 - P\left(X = 0\right) = 1 - \frac{\binom{6}{0}\binom{9}{3}}{\binom{15}{3}} = 1 - \frac{84}{455} = \frac{371}{455}$$

26. In the parent population, out of 10000, 6000 of the residents are favoring the sales tax. In the selected 15 voters, the number of residents who favor the tax is a hypergeometric random variable.

$$X \sim HypGeom\left(N = 10000, K = 6000, n = 15\right)$$

Since N and K are very large relative to n, we can apply a binomial approximation.

$$X \sim HypGeom\left(N = 10000, K = 6000, n = 15\right) \quad \approx \quad X' \sim Binom\left(n = 15, p = \frac{K}{N} = \frac{6000}{10000} = 0.6\right)$$

The question ask $P\left(X \leq 7\right)$, which can be approximated by:

$$P\left(X' \leq 7\right) = \sum_{x=0}^{7} P\left(X' = x\right) = \sum_{x=0}^{7} \binom{15}{x} \left(0.6\right)^{x} \left(0.4\right)^{15-x} \approx 0.2131$$

27. (a) The number of births that the mother bunny needs to give until the 3rd female is a negative binomial random variable:

$$X \sim NBinom\left(r = 3, p = \frac{1}{3}\right)$$

and the question asks $P\left(X = 8\right)$:

$$P\left(X = 8\right) = \binom{8-1}{3-1}\left(\frac{1}{3}\right)^{3}\left(\frac{2}{3}\right)^{8-3} = 21 \times \frac{32}{6561} \approx 0.1024$$

(b) X here is defined as in part (a). Hence $E\left[X\right] = \frac{r}{p} = \frac{3}{1/3} = 9$.

(c) Now, we know that there is a population of 9 offsprings, 6 of them being male, we choose a sample of 4 out of that. The number of male offsprings in the sample is a hypergeometric random variable.

$$X \sim HypGeom\left(N = 9, K = 6, n = 4\right)$$

The question asks the probability of $P\left(2 \leq X \leq 3\right)$.

$$\begin{aligned} P\left(2 \leq X \leq 3\right) &= P\left(X = 2\right) + P\left(X = 3\right) \\ &= \frac{\binom{6}{2}\binom{3}{2}}{\binom{9}{4}} + \frac{\binom{6}{3}\binom{3}{1}}{\binom{9}{4}} \\ &= \tfrac{45+60}{126} = \tfrac{5}{6} \end{aligned}$$

28. (a) The number of Dwight's clients in the chosen 10 is a hypergeometric random variable obviously. But we know neither the size of the parent population nor the total number of Dwight's clients. However, the question states that the number of clients is sufficiently large so we will directly apply a binomial approximation.

$$X \sim HypGeom\left(N = ?, K = ?, n = 10\right) \quad \approx \quad X' \sim Binom\left(n = 10, p = \frac{K}{N} = 0.6\right)$$

Here $p = 0.6$, as Dwight owns 60% of whole clients. Now, the question asks $P\left(X = 6\right)$:

$$P\left(X = 6\right) = \begin{pmatrix} 10 \\ 6 \end{pmatrix} (0.6)^6 (0.4)^4 \approx 0.2508$$

(b) We have 10 clients with 4 of them being James' clients. We select a sample of 5 to award with type II promotion. The number of James' clients in the sample is a hypergeometric random variable:

$$X \sim HypGeom\left(N = 10, K = 4, n = 5\right)$$

and the question asks:

$$P\left(X = 1\right) = \frac{\begin{pmatrix} 4 \\ 1 \end{pmatrix}\begin{pmatrix} 6 \\ 4 \end{pmatrix}}{\begin{pmatrix} 10 \\ 5 \end{pmatrix}} = \frac{60}{252} = \frac{5}{21}$$

(c) Again, due to large number of company's clients, we assume independence between the client promotions.

i. Out of 5 clients who are going to get type I promotion, each is likely to be James' client with $p = 0.4$ probability, as James owns 40% of the clients.

$$X \sim Binom\left(n = 5, p = 0.4\right)$$

ii. $E\left[X\right] = np = 5 \times 0.4 = 2?$

iii. $Var\left[X\right] = np\left(1 - p\right) = 5 \times 0.4 \times 0.6 = 1.2$

29. (a) Let X be the Poisson RV that represents the number of misprints on any page.

$$X \sim Pois\left(\lambda = 0.25\right) \Rightarrow P\left\{X = 0\right\} = e^{-0.25}\frac{0.25^0}{0!} \approx 0.7788$$

Let Y be the number of pages with no misprints in the next five pages.

$$Y \sim Binom\left(n = 5, p = 0.7788\right) \Rightarrow P\left\{Y = 3\right\} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} (0.7788)^3 (0.2212)^2 \approx 0.2311$$

(b) Starting from page 1, let T be the first page with at least one misprint.

$$T \sim Geom\left(p = 0.2212\right) \Rightarrow P\left\{T = 6\right\} = (0.7788)^5 (0.2212) \approx 0.0634$$

30. There are $N = 813$ elks with $K = 475$ female. Though the selection process is without replacement, since N and K is sufficiently large, we can approximate the probability of each time selecting a female with $p \approx \frac{K}{N} \approx 0.5843$. The number of elks selected until 4 females are collected is a negative binomial random variable.

$$X \sim NBinom\left(r = 4, p = 0.5843\right) \Rightarrow P\left\{X = 8\right\} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} (0.5843)^4 (0.4157)^4 \approx 0.1218$$

The exact probability calculated under the effects of non-replacement is:

$$\begin{pmatrix} 7 \\ 3 \end{pmatrix} \frac{475 \times \ldots \times 472 \times 338 \times \ldots \times 335}{813 \times 812 \times \ldots \times 806} \approx 0.1223$$

31. $X \sim Binom\left(n = 1500, p = \frac{1}{500} = 0.002\right) \approx Pois\left(\lambda = np = 3\right)$ since n is sufficiently large and p is sufficiently small. The question asks the conditional probability $P\left(X \geq 4 | X \geq 1\right)$:

$$P\left\{X \geq 4\right\} = 1 - P\left\{X = 0\right\} - P\left\{X = 1\right\} - P\left\{X = 2\right\} - P\left\{X = 3\right\} = 1 - e^{-3}\left(\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!} + \frac{3^3}{3!}\right) \approx 0.3528$$

$$P\left\{X \geq 2\right\} = 1 - P\left\{X = 0\right\} - P\left\{X = 1\right\} = 1 - e^{-3}\left(\frac{3^0}{0!} + \frac{3^1}{1!}\right) \approx 0.8009$$

Hence:

$$P\left(X \geq 4 | X \geq 2\right) = \frac{P\left(\left(X \geq 4\right) \cap \left(X \geq 2\right)\right)}{P\left(X \geq 2\right)} = \frac{P\left(X \geq 4\right)}{P\left(X \geq 1\right)} = \frac{0.3528}{0.8009} \approx 0.4405$$

32. $X \sim HypGeom\left(N = 37950, K = 12250, n = 12\right) \approx Binom\left(n = 12, p = \frac{K}{N} \approx 0.3228\right)$ since N and K are sufficiently large relative to n.

$$P\left\{X = 6\right\} = \begin{pmatrix} 12 \\ 6 \end{pmatrix} \left(0.3228\right)^6 \left(0.6772\right)^6 \approx 0.1008$$

33. Let A be the event that Pam's on an "on" day. We have $P\left(A\right) = 0.80$. Let X denote the number of shots taken to observe the third 3-pts score. X is negative binomial when the type of day is known.

$$X | A \sim NBinom\left(r = 3, p = 0.60\right), \quad X | A^c \sim NBinom\left(r = 3, p = 0.35\right)$$

(a)

$$\begin{aligned} P\left\{X = 6\right\} &= P\left(A\right) P\left\{X = 6 | A\right\} + P\left(A^c\right) P\left\{X = 6 | A^c\right\} \\ &= 0.80 \times \begin{pmatrix} 5 \\ 2 \end{pmatrix} \left(0.60\right)^3 \left(0.40\right)^3 + 0.20 \times \begin{pmatrix} 5 \\ 2 \end{pmatrix} \left(0.35\right)^3 \left(0.65\right)^3 \approx 0.1341 \end{aligned}$$

(b)

$$\begin{aligned} P\left(A^c | X = 6\right) &= \frac{P(A^c \cap (X=6))}{P(X=6)} \\ &= \frac{P(A^c)P\{X=6|A^c\}}{P(A)P\{X=6|A\}+P(A^c)P\{X=6|A^c\}} \\ &= \frac{0.20 \times \begin{pmatrix} 5 \\ 2 \end{pmatrix} (0.35)^3 (0.65)^3}{0.80 \times \begin{pmatrix} 5 \\ 2 \end{pmatrix} (0.60)^3 (0.40)^3 + 0.20 \times \begin{pmatrix} 5 \\ 2 \end{pmatrix} (0.35)^3 (0.65)^3} \\ &\approx \frac{0.0235}{0.1341} \approx 0.1752 \end{aligned}$$

34. Let Y denote the number of bullseye hits in 10 shots. We know that $Y \sim Binom\left(n = 10, p = 0.5\right)$

$$\begin{aligned} P\left(1 < X \leq 8\right) &= P\left\{X = 3\right\} + P\left\{X = 5\right\} \\ &= P\left\{3 \leq Y \leq 9\right\} \\ &= 1 - P\left\{Y = 0\right\} - P\left\{Y = 1\right\} - P\left\{Y = 2\right\} - P\left\{Y = 10\right\} \\ &= 1 - \left(0.5\right)^{10} - 10\left(0.5\right)^{10} - 45\left(0.5\right)^{10} - \left(0.5\right)^{10} = 1 - 57\left(0.5\right)^{10} \approx 0.9443 \end{aligned}$$

35. Let X be the number of flips that a student performs until a tail is observed.

$$X \sim Geom\left(p = \frac{1}{3}\right) \Rightarrow P\{X = 3\} = \frac{1}{3}\left(\frac{2}{3}\right)^2 = \frac{4}{27}$$

Let Y be the random variable that represents the number of students among 8 thatfinish this experiment in exactly 3 flips.

$$Y \sim Binom\left(n = 8, p = \frac{4}{27}\right) \Rightarrow P\{Y = 4\} = \binom{8}{4}\left(\frac{4}{27}\right)^4\left(\frac{23}{27}\right)^4 \approx 0.0178$$

36. $X \sim Binom\,(n = 1200, p = 0.002) \approx Pois\,(\lambda = np = 2.4)$ since n is sufficiently large and p is sufficiently small.

$$P\{X \geq 3\} = 1 - P\{X = 0\} - P\{X = 1\} - P\{X = 2\} = 1 - e^{-2.4}\left(\frac{(2.4)^0}{0!} + \frac{(2.4)^1}{1!} + \frac{(2.4)^2}{2!}\right) \approx 0.4303$$

37. Let X represent the number of fire incidents in any month.

$$X \sim Pois\,(\lambda = 2.4) \Rightarrow P\{X = 0\} = e^{-2.4}\frac{(2.4)^0}{0!} \approx 0.0907$$

Let Y represent the first month with no fire incident.

$$Y \sim Geom\,(p = 0.0907) \Rightarrow P\{Y = 4\} = (0.0907)\,(0.9093)^3 \approx 0.0682$$

38. Let A be the event that the randomly selected person had the shot. We have $P(A) = 0.25$. Let X denote the number of flu infections experienced in a year. X is Poisson distributed when we know the status of the person on vaccine.

$$X|A \sim Pois\,(\lambda = 0.5)\,, \quad X|A^c \sim Pois\,(\lambda = 1.5)$$

The question asks the probability $P(A|X = 1)$ which can be calculated by Bayes' theorem:

$$P(A|X = 1) = \frac{P(A \cap (X = 1))}{P(X = 1)} = \frac{P(A)\,P\{X = 1|A\}}{P(A)\,P\{X = 1|A\} + P(A^c)\,P\{X = 1|A^c\}}$$

Let us calculate the denominator:

$$\begin{aligned} P\{X = 1\} &= P(A)\,P\{X = 1|A\} + P(A^c)\,P\{X = 1|A^c\} \\ &= 0.25 \times e^{-0.5}\tfrac{0.5^1}{1!} + 0.75 \times e^{-1.5}\tfrac{1.5^1}{1!} \approx 0.3268 \end{aligned}$$

and the numerator is:

$$P(A)\,P\{X = 1|A\} = 0.25 \times e^{-0.5}\frac{0.5^1}{1!} \approx 0.0758$$

Thus, the answer is:

$$P(A|X = 1) = \frac{0.0758}{0.3268} \approx 0.2319$$

39. $X \sim Pois\,(\lambda = 3)$

$$P\{X > 1|X \geq 1\} = \tfrac{P\{(X>1)\cap(X\geq 1)\}}{P\{X\geq 1\}} = \tfrac{P\{X>1\}}{P\{X\geq 1\}} = \tfrac{1-P\{X=0\}-P\{X=1\}}{1-P\{X=0\}} = \tfrac{1-e^{-3}-3e^{-3}}{1-e^{-3}} \approx 0.8428$$

40. Let X be a hypergeometric random variable with parameters:

$$X \sim HyperGeom\left(N = 20, K = 6, n = 5\right)$$

(a) The question asks for:

$$P\left\{X = 3\right\} = \frac{\binom{6}{3}\binom{14}{2}}{\binom{20}{5}} = \frac{20 \times 91}{15504} \approx 0.1174$$

(b) The possible values that X can take are 0, 1, 2, 3, 4, and 5 with following probabilities.

$$P\left\{X = 0\right\} = \frac{\binom{6}{0}\binom{14}{5}}{\binom{20}{5}} = \frac{2002}{15504} \approx 0.1291$$

$$P\left\{X = 1\right\} = \frac{\binom{6}{1}\binom{14}{4}}{\binom{20}{5}} = \frac{6\times1001}{15504} \approx 0.3874$$

$$P\left\{X = 2\right\} = \frac{\binom{6}{2}\binom{14}{3}}{\binom{20}{5}} = \frac{15\times364}{15504} \approx 0.3522$$

$$P\left\{X = 3\right\} = \frac{\binom{6}{3}\binom{14}{2}}{\binom{20}{5}} = \frac{20\times91}{15504} \approx 0.1174$$

$$P\left\{X = 4\right\} = \frac{\binom{6}{4}\binom{14}{1}}{\binom{20}{5}} = \frac{15\times14}{15504} \approx 0.0135$$

$$P\left\{X = 5\right\} = \frac{\binom{6}{5}\binom{14}{0}}{\binom{20}{5}} = \frac{6}{15504} \approx 0.0004$$

Hence the probability mass function is:

$$p\left(x\right) = \begin{cases} 0.1291, & x = 0 \\ 0.3874, & x = 1 \\ 0.3522, & x = 2 \\ 0.1174, & x = 3 \\ 0.0135, & x = 4 \\ 0.0004, & x = 5 \\ 0, & \text{elsewhere} \end{cases}$$

and the expected value is simply $E\left[X\right] = \frac{nK}{N} = \frac{5\times6}{20} = 1.5$.